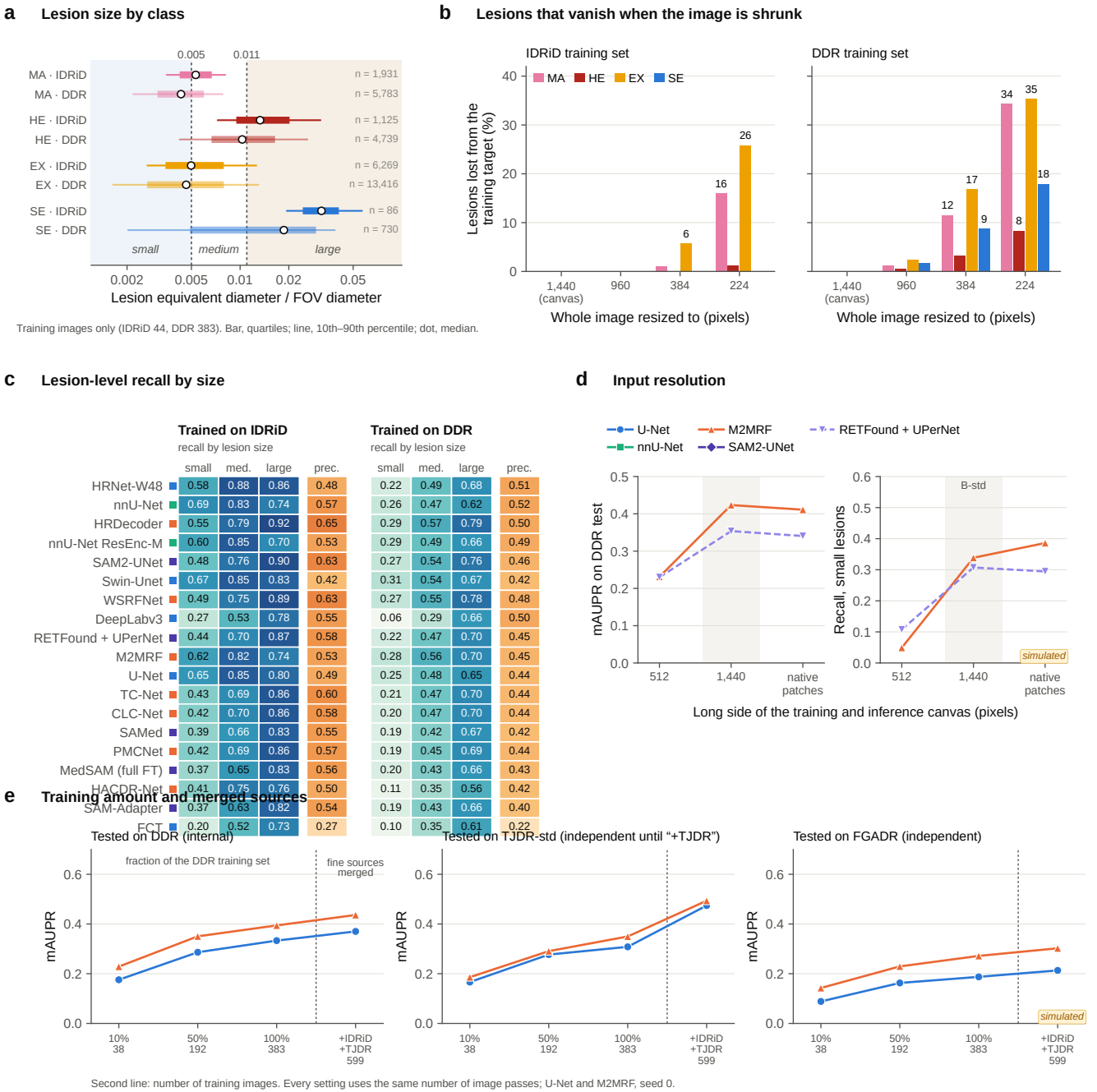
a
Lesion size by class
0.005
0.011
MA · IDRiD
MA · DDR
HE · IDRiD
HE · DDR
EX · IDRiD
EX · DDR
SE · IDRiD
SE · DDR
n = 1,931
n = 5,783
n = 1,125
n = 4,739
n = 6,269
n = 13,416
n = 86
n = 730
small
medium
large
0.002
0.005
0.01
0.02
0.05
Lesion equivalent diameter / FOV diameter
Training images only (IDRiD 44, DDR 383). Bar, quartiles; line, 10th–90th percentile; dot, median.
b
Lesions that vanish when the image is shrunk
IDRiD training set
DDR training set
MA
HE
EX
SE
Lesions lost from the training target (%)
1,440 (canvas)
960
384
224
Whole image resized to (pixels)
c
Lesion-level recall by size
Trained on IDRiD
recall by lesion size
Trained on DDR
recall by lesion size
small
med.
large
prec.
HRNet-W48 0.58 0.88 0.86 0.48 0.22 0.49 0.68 0.51
nnU-Net 0.69 0.83 0.74 0.57 0.26 0.47 0.62 0.52
HRDecoder 0.55 0.79 0.92 0.65 0.29 0.57 0.79 0.50
nnU-Net ResEnc-M 0.60 0.85 0.70 0.53 0.29 0.49 0.66 0.49
SAM2-UNet 0.48 0.76 0.90 0.63 0.27 0.54 0.76 0.46
Swin-Unet 0.67 0.85 0.83 0.42 0.31 0.54 0.67 0.42
WSRFNet 0.49 0.75 0.89 0.63 0.27 0.55 0.78 0.48
DeepLabv3 0.27 0.53 0.78 0.55 0.06 0.29 0.66 0.50
RETFound + UPerNet 0.44 0.70 0.87 0.58 0.22 0.47 0.70 0.45
M2MRF 0.62 0.82 0.74 0.53 0.28 0.56 0.70 0.45
U-Net 0.65 0.85 0.80 0.49 0.25 0.48 0.65 0.44
TC-Net 0.43 0.69 0.86 0.60 0.21 0.47 0.70 0.44
CLC-Net 0.42 0.70 0.86 0.58 0.20 0.47 0.70 0.44
SAMed 0.39 0.66 0.83 0.55 0.19 0.42 0.67 0.42
PMCNet 0.42 0.69 0.86 0.57 0.19 0.45 0.69 0.44
MedSAM (full FT) 0.37 0.65 0.83 0.56 0.20 0.43 0.66 0.43
HACDR-Net 0.41 0.75 0.76 0.50 0.11 0.35 0.56 0.42
SAM-Adapter 0.37 0.63 0.82 0.54 0.19 0.43 0.66 0.40
FCT 0.20 0.52 0.73 0.27 0.10 0.35 0.61 0.22
d
Input resolution
U-Net
M2MRF
RETFound + UPerNet
nnU-Net
SAM2-UNet
mAUPR on DDR test
Recall, small lesions
B-std
simulated
512
1,440
native patches
Long side of the training and inference canvas (pixels)
e
Training amount and merged sources
Tested on DDR (internal)
Tested on TJDR-std (independent until "+TJDR")
Tested on FGADR (independent)
fraction of the DDR training set
fine sources merged
mAUPR
10% 38
50% 192
100% 383
+IDRiD +TJDR 599
simulated
Second line: number of training images. Every setting uses the same number of image passes; U-Net and M2MRF, seed 0.